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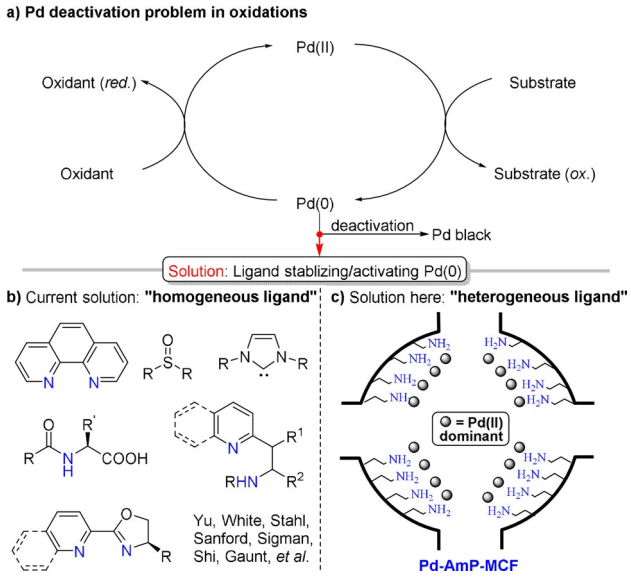
Efficient Heterogeneous Palladium-Catalyzed Oxidative Cascade Reactions of Enallenols to Furan and Oxaborole Derivatives

Man-Bo Li, Daniels Posevins, Antoine Geoffroy, Can Zhu, and Jan-E. Bäckvall**

Abstract: *A heterogeneous palladium-catalyzed oxidative cyclization of enallenols has been developed for the construction of highly substituted furan and oxaborole derivatives. The heterogeneous catalyst (Pd-AmP-MCF) exhibits high activity, high site- and stereoselectivity, and efficient palladium recyclability in the transformations.*

Pd-catalyzed oxidations[1] such as the Wacker process,[2] alkene functionalizations,[3] and C–H activations[4] constitute an important research field and these reactions have become increasingly important in organic synthesis, as they enable the diverse functionalizations of unsaturated C–C bonds and unreactive C–H bonds. In spite of the significant progress in homogeneous Pd-catalyzed oxidations, the achievement of high efficiency is still challenging in these reactions[5] due to the generation of palladium black from active palladium species during the catalytic cycle (Scheme 1 a). To circumvent the Pd deactivation problem in homogeneous oxidation reactions, considerable efforts have been made by different groups to develop ligands that are stable towards the oxidizing conditions and that can keep Pd^0 in solution (Scheme 1 b).[6] Apart from solving the Pd deactivation problem shown in Scheme 1 a, control of selectivity and efficient recyclability of the precious palladium catalyst are other challenges in Pd-catalyzed oxidations.[7]

Our group has a long-standing interest in Pd-catalyzed oxidations.[1a,b,8] Recently, we reported on a heterogeneous nanopalladium catalyst, where palladium is immobilized into the cavities of aminopropyl-functionalized siliceous mesocellular foam (Pd-AmP-MCF).[9a] The performance of this catalyst has been preliminary tested in organic transformations, such as dynamic kinetic resolution, hydrogenation, and oxidation.[9] This catalyst has shown high activity as well as excellent recyclability. Inspired by these results, we present herein the work on the use of Pd-AmP-MCF as a general and efficient heterogeneous catalyst in Pd-catalyzed oxidative cyclizations of enallenols via $C(sp^3)$–H oxidations. This strategy provides a new and satisfactory solution of the Pd deactivation problem (Scheme 1 c), leading to high Pd efficiency, high site- and stereoselectivity and palladium recyclability. As far as we known, there are only a few reports on the heterogeneous Pd-catalyzed $C(sp^3)$–H oxidations.[9c,d,10]



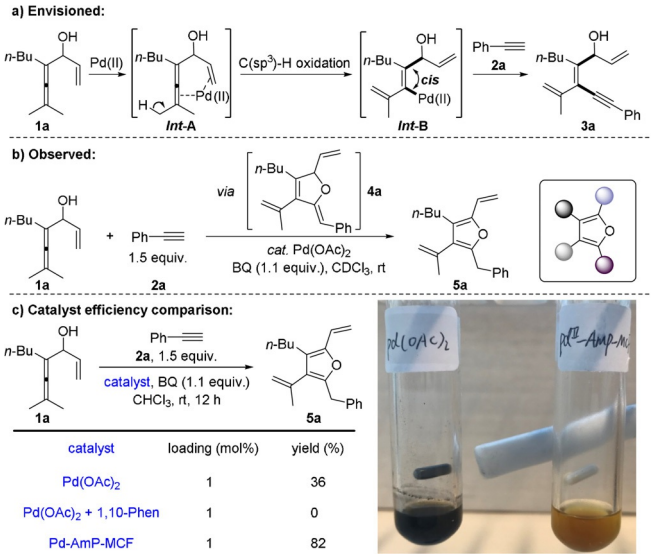
Scheme 1. Pd deactivation problem in oxidations and the solutions.

We initiated our study by using enallenol **1a** and alkyne **2a** as the model substrates. We envisioned that the simultaneous coordination (***Int*-A**) of the allene and olefin unit of **1a** would trigger the allene attack[8c] via $C(sp^3)$–H bond oxidation to form the *cis*-palladium intermediate (***Int*-B**) with the Pd^{II} and the allylic alcohol group on the same side of the C=C bond (Scheme 2 a). In the presence of the terminal alkyne **2a**, ligand exchange and reductive elimination would produce the (*Z*)-tetrasubstituted olefin **3a**. Interestingly, treatment of **1a** and **2a** with $Pd(OAc)_2$ and *p*-benzoquinone (BQ) in $CDCl_3$ at room temperature did not give **3a**. Instead a tetrasubstituted furan **5a** was obtained as the exclusive product, which might be generated from the isomerization of **4a** (Scheme 2 b). In a previous report, we found a similar transformation of allenic amides without the pending olefin by using homogeneous palladium. However, in that case the reactivity of the corresponding allenic alcohol did not give the corresponding tetrasubstituted furan (cf. **5a**) under homogeneous conditions.[11]

Catalyst screening showed distinct differences between the homogeneous palladium and the heterogeneous Pd-AmP-MCF catalysts concerning catalyst efficiency (for more details

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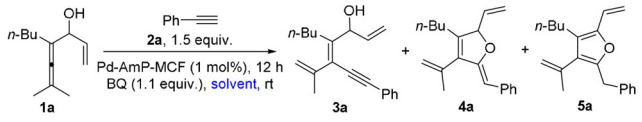


Scheme 2. a) Envisioned Pd-catalyzed C(sp^3)–H oxidative alkynylation of enallenol. b) Observed results. c) The catalyst efficiency comparison.

on catalyst screening, see Supporting Information, page S13). As shown in Scheme 2c, 1 mol% of $Pd(OAc)_2$ only gave **5a** in 36% yield at room temperature in 12 h. The addition of 1,10-phenanthroline to $Pd(OAc)_2$ completely shut down the reaction, probably due to the strong coordination of 1,10-phenanthroline to Pd^{II} disfavoring the formation of ***Int*-A** and the subsequent allene attack. Replacement of $Pd(OAc)_2$ by the heterogeneous Pd-AmP-MCF (1 mol%) gave **5a** in 82% yield.

In the homogeneous reaction with $Pd(OAc)_2$, Pd black was observed in the reaction mixture after completion, while there was no observed generation of Pd black in the reaction using the heterogeneous Pd-AmP-MCF (Scheme 2c). These results demonstrate the high efficiency of the Pd-AmP-MCF catalyst in Pd-catalyzed C(sp^3)–H oxidations, which is credited to the protection of Pd species from aggregating to Pd black.

Further solvent screening showed interesting results: by using CH_3CN, CH_3OH, or $CHCl_3$ as the solvent, (*Z*)-tetrasubstituted olefin **3a**, 2,5-dihydrofuran **4a**, or tetrasubstituted furan **5a** was obtained as the dominant product, respectively (Scheme 3, for more details on the solvent effect, see Supporting Information, P. S13).



solvent	yield of **3a** (%)	yield of **4a** (%)	yield of **5a** (%)
CH_3CN	90	0	0
CH_3OH	0	75	8
$CHCl_3$	0	0	82

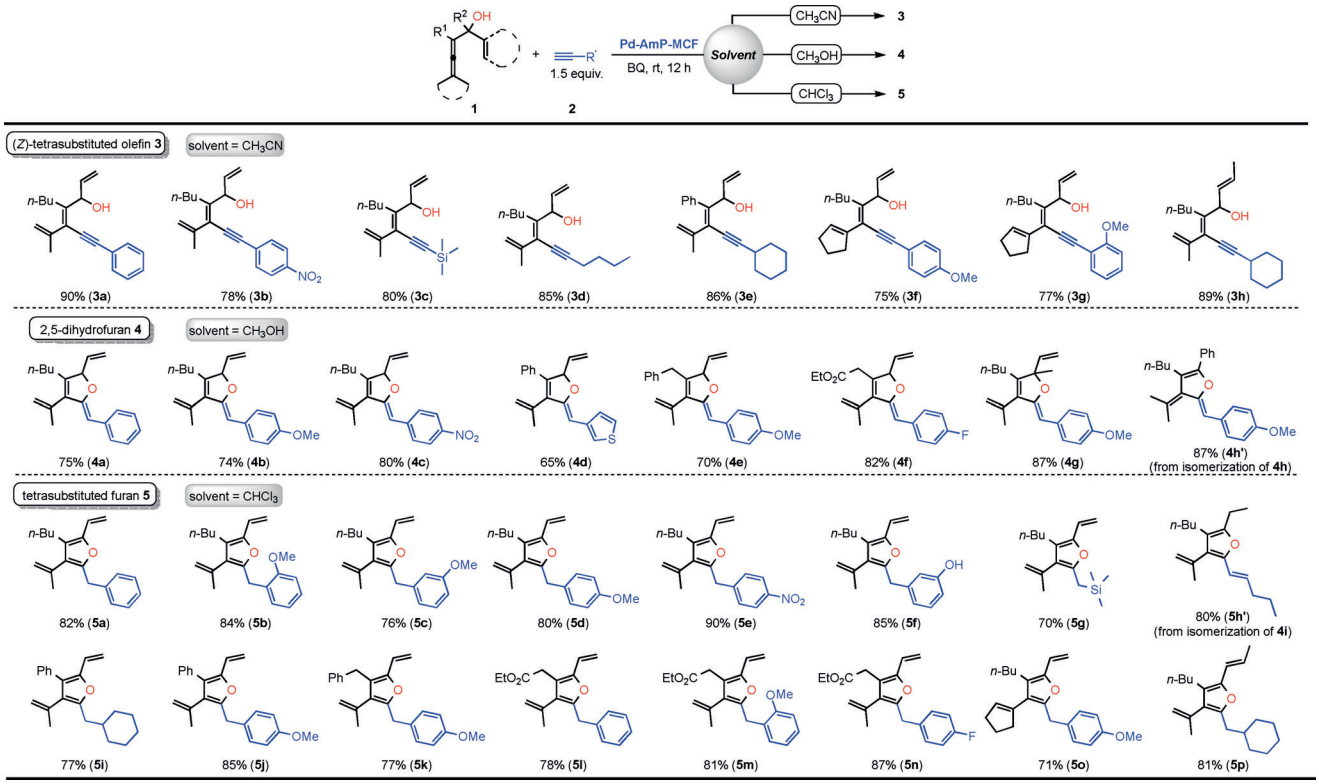
Scheme 3. Solvent-controlled chemoselectivity.

With the optimized reaction conditions in hand, we explored the substrate scope of Pd-AmP-MCF-catalyzed C(sp^3)–H oxidations of enallenols **1** for the divergent syntheses of **3**, **4**, and **5** (Scheme 4), which are important building blocks and core structures of a large variety of natural and bioactive compounds.[12] By using CH_3CN or CH_3OH as the solvent, enallenols **1** worked well with terminal alkynes **2** to give (*Z*)-tetrasubstituted olefins **3** or dihydrofurans **4**, respectively. Enallenol **1ab** with one more carbon in the pending olefin (for the structure of **1ab**, see Ref. [13]) also reacted with an alkyne in CH_3OH to give the corresponding dihydrofuran **4ab** in 80% yield.[13] The chemoselectivity of the Pd-AmP-MCF-catalyzed oxidative cyclization of enallenols **1** is simply controlled by using $CHCl_3$ as the solvent to give tetrasubstituted furan **5**. To be noted, **4h′** and **5h′** were isolated as the final products in Scheme 4, which are generated from olefin isomerization of the initially formed **4h** and **4i**, respectively (for the formation of **4h′** and **5h′**, see Ref. [14]).[14] Pd leaching was not observed in the heterogeneous catalysis based on a hot filtration test, which rules out a reaction via a homogeneous pathway. Kinetic studies (Figure 1) revealed that the rate and yield of the reaction are essentially maintained between the 1st and the 6th runs, which demonstrates that Pd-Amp-MCF is robust and has maintained its activity in 6 runs. However, we cannot exclude the possible loss of activity in the long run of the heterogeneous catalyst.[15]

We compared the X-ray photoelectron spectroscopy (XPS) and transmission electron microscopy (TEM) images of Pd-AmP-MCF before reaction and after six cycles. The deconvoluted Pd3d XPS spectrum of Pd-AmP-MCF before reaction showed intensive Pd^{II} signals at 338 eV and 343 eV (Figure 2a). After 6 cycles, the catalyst still had a dominant proportion of Pd^{II} (Figure 2b). TEM images showed that no palladium black was generated from Pd species of Pd-AmP-MCF after 6 cycles (Figure 2c and d). Based on these results, we conclude that the porous support (AmP-MCF) of Pd-AmP-MCF protect Pd species from aggregating to catalytically inert Pd black during the catalytic cycle, which leads to the high efficiency of Pd-AmP-MCF.

To gain further insight into the reaction mechanism, we carried out control experiments (Scheme 5). First, we replaced the vinyl group of enallenol **1a** by an ethyl group to form **1ac**. Treatment of allenol **1ac** with terminal alkyne **2a** under standard reaction conditions in different solvents did not afford the corresponding products, instead **1ac** was recovered in >90% yield in these cases (Scheme 5a). These results show that the olefin group of enallenols **1** is required for coordinating to palladium to initiate the C(sp^3)–H oxidation and control the stereoselectivity, furnishing a *cis*-palladium intermediate (***Int*-B** in Scheme 2a). Second, **3a** was treated with 1 mol% of Pd-AmP-MCF in CH_3OH or $CHCl_3$, which afforded **4a** or **5a**, respectively (Scheme 5b). Furthermore, **4a** was transformed into **5a** in $CHCl_3$ in good yield. These results demonstrate that 2,5-dihydrofuran **4a** is formed via Pd-catalyzed intramolecular cyclization of **3a**, and that tetrasubstituted furan **5a** is formed via isomerization of **4a**. Based on the experimental results, we propose a mechanism for the Pd-AmP-MCF-catalyzed oxidative functionalizations

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Scheme 4. Substrate scope and solvent-controlled selectivity of the Pd-AmP-MCF-catalyzed C(sp^3)–H oxidation of enallenols **1** for the divergent construction of (*Z*)-tetrasubstituted olefin **3**, dihydrofuran **4**, and tetrasubstituted furan **5**.

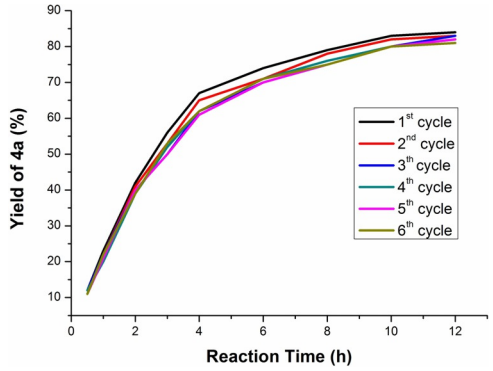


Figure 1. Recycling of Pd-AmP-MCF in the transformation from **1a** to **5a**.

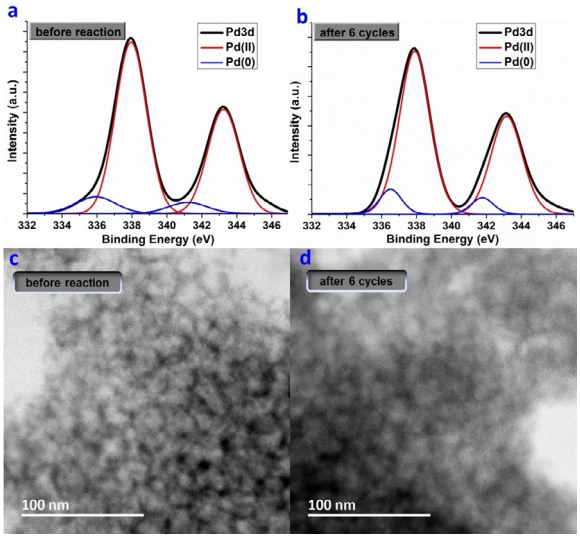


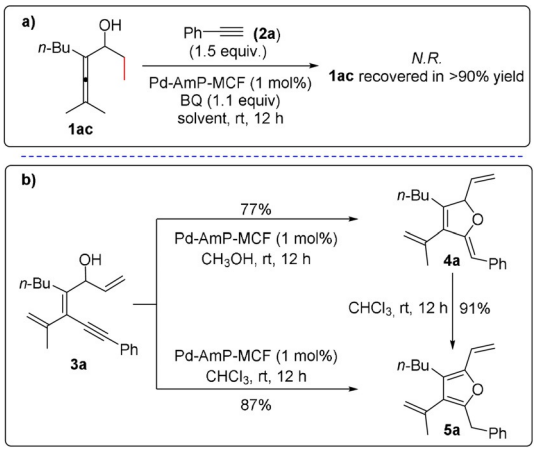
Figure 2. XPS and TEM analyses of Pd-AmP-MCF before reaction and after six cycles.

of enallenols, leading to (*Z*)-tetrasubstituted olefin **3**, 2,5-dihydrofuran **4**, or tetrasubstituted furan **5** (Scheme 6).

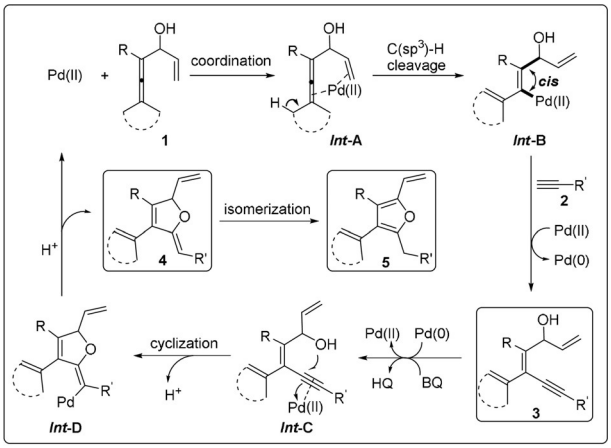
On the basis of mechanistic understanding of the Pd-AmP-MCF-catalyzed oxidative cyclizations of enallenols **1**, we extended this strategy for the construction of oxaborole **6**, which is an important synthon in organic transformations.[16] As reported by Baran,[16a] this oxaborole skeleton is a key intermediate permitting rapid and gram-scale access to a variety of bioactive meroterpenoids. By replacing the alkyne reaction partner by bis(pinacolato)diboron (B_2pin_2), in the presence of 1 mol% of Pd-AmP-MCF and 1.1 equiv of BQ in $CHCl_3$ at room temperature, enallenols **1** were transformed to oxaboroles **6** in excellent yields (Scheme 7).[17]

More importantly, by using enantiomerically pure enallenols **1**, readily obtained from kinetic resolution of enallenols **1** with *Candida antarctica* lipase B (CalB), the reaction afforded optically pure (> 99% *ee*) oxaboroles **6** in high yields (for the preparation of enantiomerically pure enallenols **1**, see Supporting Information, Page S41). During the process, no loss of optical purity was observed, in spite of the potential racemization of the allylic alcohol by palladium. This hetero-

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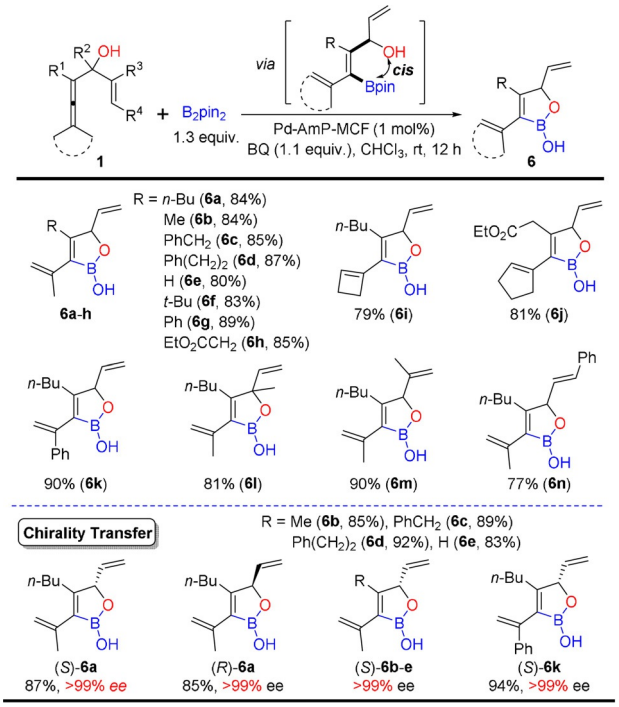
Scheme 5. Control experiments.



Scheme 6. Proposed mechanism.

geneous process provides an efficient route towards highly substituted chiral oxaboroles with high selectivity and step-economy. This method should find use in medicinal chemistry as a result of the wide application of oxaboroles in antibacterials[18] and their rapid and divergent access to bioactive meroterpenoids.[16a]

In conclusion, we have developed a heterogeneous palladium-catalyzed oxidative cascade cyclization of enallenols, constructing highly substituted furan and oxaborole derivatives. The heterogeneous process circumvents the Pd deactivation problem in homogeneous Pd-catalyzed oxidations, exhibing high catalyst efficiency, high site- and stereoselectivity and efficient palladium recyclability. It was demonstrated that the support (AmP-MCF) of the heterogeneous catalyst (Pd-AmP-MCF) protects the Pd species from aggregating to Pd black during the catalytic cycle, which leads to the high Pd efficiency of the heterogeneous palladium catalyst. The Pd-AmP-MCF is emerging as an important replacement for homogeneous palladium catalysts, such as palladium acetate, in selective oxidation reactions, and further studies on its use in various Pd-catalyzed oxidative transformations are currently underway in our laboratory.



Scheme 7. Pd-AmP-MCF-catalyzed oxidative cyclization of enallenols for the construction of oxaboroles.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: efficiency · heterogeneous palladium · oxidative cyclizations · stereospecific

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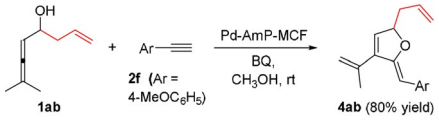
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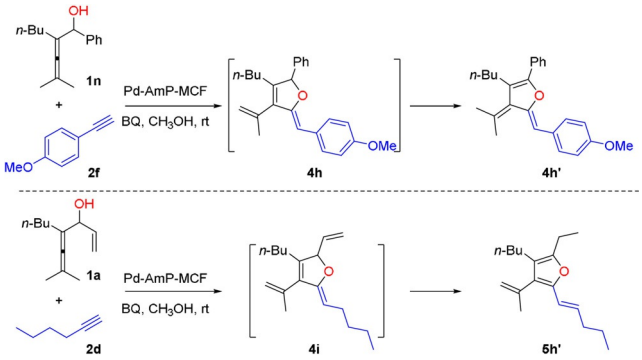
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[13] For more information on the reaction of **1ab** to **4ab** (below), see Supporting Information, P. S33.



[14] Formation of **4h′** and **5h′**. (for a possible isomerization pathway for the formation of **5h′**, see Supporting Information, P. S29)



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[17] The transformation of enallenol **1** is switched to give cyclobutenol as the major product by using MeOH as the solvent and 0.1 equiv of Et_3N as the additive (Ref. [9c]), which was explained by the interaction between triethylamine and the palladium catalyst (e.g., coordination) favoring olefin insertion from ***Int*-B**. The lack of triethylamine would favor the direct transmetallation of ***Int*-B**, and the relatively acidic solvent $CHCl_3$ would favor the alcoholysis of the *cis* vinyl intermediate (in Scheme 7), forming oxaboroles **6**.

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